

Characteristic Witnesses at Limits and Their First Successors for the Extended Oscillation Rank

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Status

Upgraded partial result, likely valid; human verification recommended. The source asks whether every nonzero countable ordinal is an exact value of β_ξ^* . This packet proves characteristic realizability at every nonzero countable limit and at the first successor of every such limit. In combination with the predecessor packet, exact values now include

$$1, \quad 2, \quad \lambda, \quad \lambda + 1 \quad (0 < \lambda < \omega_1 \text{ a limit}).$$

Finite ranks at least 3, and finite tails $\lambda + n$ with $n \geq 2$, are not settled here.

Source Question

Leung, Ng, and Tang ask after Proposition 3.6 of [1]:

Remark. In an uncountable Polish space (X, τ) , any nonzero countable ordinal is in the range of $\beta = \beta_1^*$. Propositions 3.4 and 3.6 say that this is “almost” true for $\xi \geq 2$. It would be interesting to find out if it is true exactly.

Question. Let (X, τ) be an uncountable Polish space and let ξ be a countable ordinal ≥ 2 . Is it true that for any nonzero countable ordinal ζ , there exists $f \in \mathfrak{B}_\xi(\tau)$ such that $\beta_\xi^*(f) = \zeta$?

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The source proves exact realizability of every nonzero countable limit by a not necessarily characteristic function [1, Proposition 3.6].

Difference and Potential Ranks

For disjoint $\Pi_{\xi+1}^0$ sets A, B , let $\alpha_\xi(A, B)$ be the least length of a decreasing continuous sequence of Π_ξ^0 sets whose alternating difference contains A and is disjoint from B . For a Baire class ξ function f , $\alpha_\xi(f)$ is the supremum of these separation lengths over the rational level pairs $\{f \leq p\}, \{f \geq q\}$.

Let $\alpha_\xi^*(f)$ be the minimum ordinary separation rank over the same admissible Polish refinements used to define $\beta_\xi^*(f)$. Elekes, Kiss, and Vidnyánszky prove the exact inequalities

$$\alpha_\xi^*(f) \leq \alpha_\xi(f) \leq 2\alpha_\xi^*(f), \quad (1)$$

where multiplication is ordinal multiplication from the left [2, Theorem 5.17].

We use the classical strictness theorem for the Hausdorff difference hierarchy: on Cantor space, for every $1 \leq \xi < \omega_1$ and every nonzero countable θ , there is a $K \in \mathbf{\Delta}_{\xi+1}^0$ whose least continuous alternating-difference length over $\mathbf{\Pi}_\xi^0$ is exactly θ ; see [3, Section 22]. This is the main external descriptive-set-theoretic input.

Lemma 1 (Characteristic functions). *For every admissible Polish topology σ and every set H for which χ_H is Baire class one in σ ,*

$$\alpha_\sigma(\chi_H) = \beta_\sigma(\chi_H).$$

Consequently $\alpha_\xi^(\chi_H) = \beta_\xi^*(\chi_H)$.*

Proof. For the only nontrivial rational level pairs of χ_H , the two separated sets are H^c and H . Their separation derivative on a closed set F is

$$\overline{F \cap H}^\sigma \cap \overline{F \cap H^c}^\sigma.$$

For every $0 < \varepsilon \leq 1$, this is also exactly the oscillation derivative $D_\sigma(\chi_H, \varepsilon, F)$. For $\varepsilon > 1$ the latter is empty. The transfinite iterates, their ranks, and then the minima over admissible topologies therefore agree. $\square$

Exact Realization Theorem

Theorem 1. *Let (X, τ) be an uncountable Polish space and let $2 \leq \xi < \omega_1$. Suppose that θ is either*

1. *a nonzero countable limit ordinal, or*
2. *$\lambda + 1$, where λ is a nonzero countable limit ordinal.*

Then there is a set $A \subseteq X$ such that $\chi_A \in \mathfrak{B}_\xi(X, \tau)$ and

$$\beta_\xi^*(\chi_A) = \theta.$$

Proof. Every uncountable Polish space contains a closed subset P homeomorphic to Cantor space. Work first in P . By strictness of the difference hierarchy, choose $K \in \mathbf{\Delta}_{\xi+1}^0(P)$ of exact $\mathbf{\Pi}_\xi^0(P)$ -difference length θ , and put $H = P \setminus K$.

For $f = \chi_H$ the only nontrivial separation problem is the ordered pair (K, H) . A set separating K from $H = P \setminus K$ must equal K . Therefore the orientation is fixed and

$$\alpha_{\xi, P}(f) = \theta. \quad (2)$$

Since $H, K \in \mathbf{\Delta}_{\xi+1}^0(P)$, the characteristic function f is of Baire class at most ξ .

Set

$$r = \beta_{\xi, P}^*(f) = \alpha_{\xi, P}^*(f),$$

using Lemma 1. Equations (1) and (2) give

$$r \leq \theta \leq 2r. \quad (3)$$

If θ is a limit and $r < \theta$, then

$$2r < r + \omega \leq \theta.$$

This contradicts $\theta \leq 2r$ in (3). Hence $r = \theta$.

Now let $\theta = \lambda + 1$, with λ a nonzero limit. If $r < \lambda + 1$, then $r \leq \lambda$, so

$$2r \leq 2\lambda = \lambda < \lambda + 1 = \theta,$$

again contradicting (3). The identity $2\lambda = \lambda$ follows from $2\eta < \eta + \omega \leq \lambda$ for every $\eta < \lambda$ and continuity of ordinal multiplication in its right argument. Thus in both cases

$$\beta_{\xi, P}^*(\chi_H) = \theta. \quad (4)$$

Finally extend χ_H by zero from P to X , equivalently regard $H \subseteq P \subseteq X$ and use $g = \chi_H$ on X . Lemma 3.5 of [1] gives

$$\theta = \beta_{\xi, P}^*(g|_P) \leq \beta_{\xi, X}^*(g) \leq 1 + \beta_{\xi, P}^*(g|_P) = 1 + \theta.$$

If θ is a nonzero limit, $1 + \theta = \theta$. If $\theta = \lambda + 1$, then $1 + \theta = (1 + \lambda) + 1 = \lambda + 1 = \theta$. Hence the two bounds agree. The same lemma ensures that the zero extension is Baire class ξ on X , and the proof is complete. $\square$

Adversarial Verification

Step	Verdict	Check
Exact difference level	External, standard	Strictness must provide exact least continuous Π_ξ^0 -difference length θ on Cantor space, not merely unboundedness.
Orientation	Valid	For $f = \chi_H$, the low/high pair is (K, H) ; because these are complements in P , its only separator is K .
$\alpha^* = \beta^*$	Valid	The ordinary separation and oscillation derivatives are literally the same for an indicator.
Factor-two lower bound	Valid	Theorem 5.17 is an exact ordinal inequality, not only essential equivalence.
Limit arithmetic	Valid	For $r < \theta$ with θ limit, $r + \omega \leq \theta$; and for limit λ , $2\lambda = \lambda$.
Closed transfer	Valid	Lemma 3.5 sandwiches the ambient rank between θ and $1 + \theta$, which equals θ in both cases used.
Scope	Partial	No argument covers $\lambda + n$ for $n \geq 2$ or finite ranks ≥ 3 .

Why the Argument Stops at the Next Finite Tail

For $\theta = \lambda + n$ with $n \geq 2$, a hypothetical strict drop to $r = \lambda + n - 1$ is compatible with the comparison theorem because

$$2r = \lambda + 2n - 2 \geq \lambda + n = \theta.$$

Thus exact difference length alone no longer prevents a lower potential rank. A full proof would need a calibrated witness that simultaneously has an admissible upper topology of rank $\lambda + n$ and a stronger refinement-resistant lower invariant. A one-point successor gadget is destroyed by making the point clopen, while a tested finite rooted-tree calibration did not attain the required sharp factor two. Neither route is promoted as a theorem.

Scope and Novelty Check

Together with the rank-two packet, Theorem 1 gives exact characteristic witnesses for 1, 2, every nonzero countable limit, and the first successor of every nonzero countable limit. This strengthens the source’s limit realization by making those witnesses characteristic, but it does not answer the source question in full.

The bounded search covered the run registry and solution/attempt/proof-gap indexes; the full sources of arXiv:1701.05649 and arXiv:1406.5724; exact question phrases and rank/range searches; and citation metadata for the source paper. No explicit later full solution or first-successor statement was found. This is not an exhaustive literature review, so novelty confidence is moderate.

Human Review Recommendation

Send this packet for verification as a substantial partial result. The reviewer should focus first on the exact-level version of difference-hierarchy strictness and its continuous-sequence convention, then on the orientation in (2). Do not label the original question solved unless a new argument also covers all remaining finite tails.

References

- [1] D. H. Leung, H.-W. Ng, and W.-K. Tang, *On ordinal ranks of Baire class functions*, Fundamenta Mathematicae 247 (2019), 109–130; arXiv:1701.05649.
- [2] M. Elekes, V. Kiss, and Z. Vidnyánszky, *Ranks on the Baire class ξ functions*, Transactions of the American Mathematical Society 368 (2016), 8111–8143; arXiv:1406.5724.
- [3] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.